

The Ordinal Summit

Where the Tower Snaps to c_1 | Version 1.1

ZP Companion | May 2026

This companion explains the ideas in plain language with diagrams and examples. It is not the formal document — every claim here restates a result already proved in ZP-L Incomputability Convergence v1.0. Consult that document for the authoritative mathematics.

What Is ZP-L Doing?

ZP-K proved that the initial machine state c_0 is a Kleene fixed point: a program whose behavior is determined entirely by its own index, guaranteed to exist by the second recursion theorem. ZP-L picks up the thread from a different direction: the ordinal direction. If a map ϕ assigns machine phases to ordinals — sending c_0 to everything below a certain point and c_1 above it — what is that threshold ordinal? ZP-L establishes the answer: ϵ_0 , the first fixed point of the map $\alpha \mapsto \omega^\alpha$.

This is not an arbitrary choice. ZP-L proves that ϵ_0 is the minimal threshold — no monotone, tower-aligned map can snap before ϵ_0 , and the canonical map snaps exactly there. ZP-L also establishes the connection to ZP-B: as the tower stages approach ϵ_0 in ordinal order, their 2-adic encodings converge to $0 = \perp$ in $\mathbb{Z}_2$. The same sequence witnesses both convergences simultaneously.

Real-world analogy — the unreachable horizon

Imagine a path where each step is longer than the last: you walk 1 mile, then 2, then 4, then 8, ... You are always getting further from the start, but you are also always somewhere on the path — never at the horizon. The horizon (ϵ_0) is the limit of all those steps combined. It exists as a mathematical object, but no finite number of steps reaches it.

The ordinal tower works the same way. The snap that ZP-E proved occurs at ϵ_0 is not a step you can take from inside the tower. It is the point where the tower stops and something new begins.

What Is ϵ_0 ?

ϵ_0 (epsilon-zero) is the smallest ordinal α satisfying the equation $\omega^\alpha = \alpha$. In other words, it is the first fixed point of the map that sends any ordinal α to ω^α (omega raised to the power α). This is not a coincidence or a definition by fiat — it is a property ϵ_0 satisfies and no smaller ordinal does.

The tower $0, 1, \omega, \omega^\omega, \omega^{\omega^\omega}, \dots$ climbs through the ordinals, each stage being ω raised to the previous stage. Every stage is strictly below ϵ_0 , and ϵ_0 is their limit — the supremum of the whole sequence. It is never reached from below; it is the boundary above all stages.



Each tower stage is strictly below ε_0 and maps to c_0 . The snap to c_1 occurs exactly at ε_0 .

The tower stages $0, 1, \omega, \omega^\omega, \dots$ all lie strictly below ε_0 and map to c_0 . The snap transition to c_1 occurs at the threshold ε_0 itself. The arrow marks the snap. ZP-L §VII: `epsilon_zero_snap_canonical`.

Remember: ε_0 is not "infinity." It is a specific countable ordinal — the first one satisfying $\omega^\alpha = \alpha$. Ordinals larger than ε_0 exist ($\varepsilon_0 + 1, \varepsilon_1$, and many more). ZP-L is not claiming that ε_0 is the largest ordinal or that the framework breaks down above it. It is the minimal snap threshold for monotone, tower-aligned maps — and that minimality is exactly what is proved.

Why ε_0 Is the Exact Threshold

Two facts together pin ε_0 as the minimal snap threshold. They work like a lower bound and an upper bound that meet at the same point.

Lower bound (snap cannot happen before ε_0): The fundamental sequence is cofinal in ε_0 . This means: for any ordinal α below ε_0 , some tower stage eventually exceeds α . A monotone map that assigns c_0 to every tower stage must therefore assign c_0 to α too (it cannot increase between a stage above α and α itself). So no ordinal strictly below ε_0 can be a snap point for such a map.

Upper bound (snap does happen at ε_0): ε_0 is itself a fixed point of $\omega^\cdot$ ($\omega^\varepsilon_0 = \varepsilon_0$). A map that assigns c_1 to all fixed points of $\omega^\cdot$ must assign c_1 to ε_0 . Together with the lower bound, ε_0 is the first place where c_1 can and must appear.

The canonical witness map $\phi(\alpha) = c_0$ if $\alpha < \varepsilon_0$, else c_1 satisfies all five conditions at once — monotonicity, tower-alignment, fixed-point-respecting, snapping at ε_0 , and minimality — with no free hypotheses. This is the central result of ZP-L.

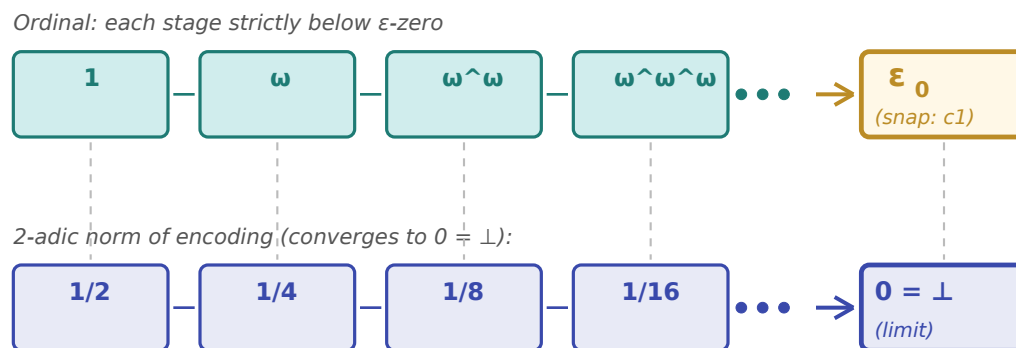
"Minimal" does not mean "unique." ZP-L proves that no snap can occur strictly before ε_0 (under the stated conditions), not that ε_0 is the only possible snap point. Maps that also assign c_1 to ordinals above ε_0 are not ruled out. The framework fixes ε_0 as the threshold by the canonical witness, not by uniqueness of all satisfying maps.

The Two-Adic Connection

ZP-B established that $0 = \perp$ in the 2-adic integers $\mathbb{Z}_2$ is the valiative sink: the 2-adic valuation of 0 is infinite, which means 0 sits at the bottom of the 2-adic metric. ZP-L connects this to the ordinal tower via an

encoding of ordinals below ε_0 into $\mathbb{Z}_2$ using their Cantor normal forms.

The encoding (cnfToZp2 in ZPL.lean) maps the n -th tower stage to 2^n in $\mathbb{Z}_2$. The 2-adic norm of 2^n is $(1/2)^n$, which goes to 0 as n increases. So as the ordinal tower climbs toward ε_0 from below, its 2-adic image converges to $0 = \perp$ in the 2-adic metric (the norms $(1/2)^n$ decrease monotonically and converge to 0). Two convergences, one tower.



Same tower, two perspectives. Teal (top): ordinal stages approach ε -zero. Indigo (bottom): their 2-adic norms approach 0.

Same tower stages, two perspectives. Teal (top row): ordinal stages $1, \omega, \omega^\omega, \dots$ approach ε_0 from below. Indigo (bottom row): the 2-adic norms of their encodings $1/2, 1/4, 1/8, \dots$ converge to $0 = \perp$. ZP-L §VII: snap_zp2_correspondence.

This dual convergence is what snap_zp2_correspondence formalizes in ZP-L §VII: four jointly established facts about the same tower sequence — all stages below ε_0 in ordinal order, all assigned c_0 by the canonical map, all encoding to 2^n in $\mathbb{Z}_2$ with 2-adic norm $(1/2)^n \rightarrow 0$, and the canonical map snapping to c_1 at ε_0 .

The Kleene Connection

ZP-K identified $\perp$ as a Kleene fixed point: a program that is its own output, guaranteed to exist by Kleene's second recursion theorem. ZP-L identifies ε_0 as an ordinal fixed point: the first ordinal satisfying $\omega^\alpha = \alpha$, guaranteed to exist by the fixed-point theorem for normal functions. Both fixed-point proofs require Classical.choice in their Lean formalizations. In each case, the choice axiom enters at a non-constructive selection step: the ordinal proof uses it for the supremum that defines ε_0 ; the Kleene proof uses it for the fixed-point selection. Whether these reflect the same underlying mathematical phenomenon — or are coincidentally parallel — is the open question ZPL.lean §I poses.

ZP-L §VI (kleene_ordinal_snap_bridge) names this structural parallel explicitly. The theorem itself is purely ordinal — no Code or eval object appears in it. What the section establishes is that the hypothesis structure of the two proofs is the same: diagonalization, non-constructive selection, fixed-point existence.

Two rooms, same pattern

Imagine two mathematicians working in separate rooms. One (ZP-K) is computing with programs and asking: what program runs itself? The other (ZP-L) is computing with ordinals and asking: what ordinal satisfies $\omega^\alpha = \alpha$? Neither knows what the other is doing. When they compare notes, they find the same proof structure: the same diagonal argument, the same non-constructive selection step, the same kind of fixed-point existence result.

ZPM formalizes this parallel with a type bridge between the two settings. The two rooms share the same floor plan.

A Note on Classical.choice

Every ZP-L theorem carries the axiom footprint `[propext, Classical.choice, Quot.sound]`. This is the standard footprint of Lean 4 + Mathlib for any theorem that uses ordinal theory, p-adic analysis, or computability. It is not a novel ZP-L commitment.

What is notable is where `Classical.choice` appears. Across the four mathematical settings of the ZP framework — topology (ZP-B), information theory (ZP-C), set theory and computation (ZP-J/K), and ordinal theory (ZP-L) — the same axiom appears at the same structural step: the non-constructive diagonal. ZP-L §I documents this convergence. Whether `Classical.choice` is structurally forced by the ZP framework — rather than merely inherited from Mathlib infrastructure — remains an open question.

Key Result — ZP-L: The Ordinal Snap Threshold

`epsilon_zero_snap_canonical` (ZPL.lean §VII): The canonical map $\phi(\alpha) = c_0$ if $\alpha < \varepsilon_0$, else c_1 satisfies all five conditions simultaneously — monotone, tower-aligned (c_0 on every fundamental sequence stage), fixed-point-respecting (c_1 on every fixed point of $\omega^\cdot$), snapping at ε_0 , and ε_0 minimal — with no free hypotheses. `snap_zp2_correspondence` co-proves that the same tower witnesses both the ordinal approach to ε_0 and the 2-adic convergence to $0 = \perp$. All theorems proved without sorry. ZPM formalizes the type bridge between the ordinal and machine-phase encodings. ✓